



CIS 241

System-level Programming and Utilities

Welcome! :^)

Instructor: Dr. Austin Ferguson

Pronouns: he/him

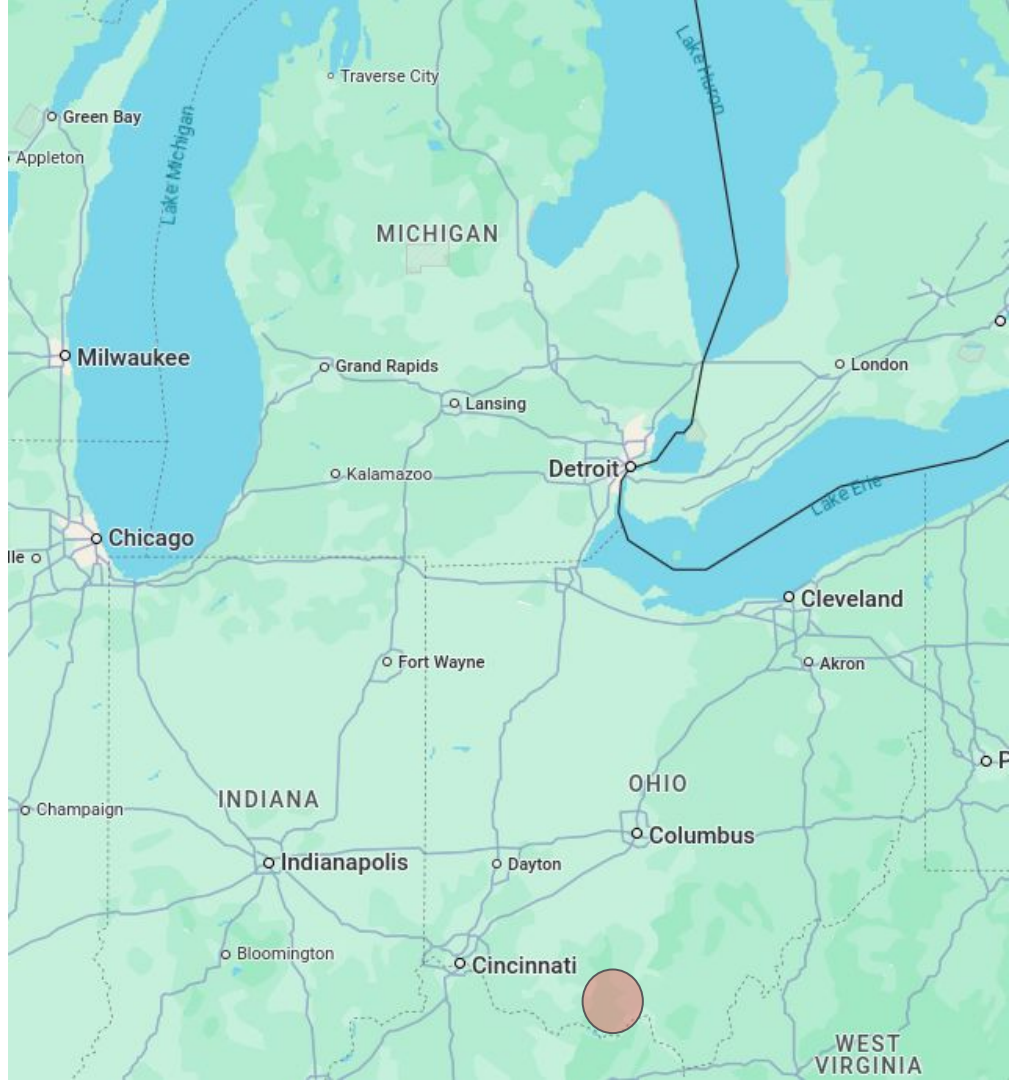
Email: ferguaus@gvsu.edu

Office: MAK C-2-216

Adapted from materials from Dr. Carrier

Who am I?

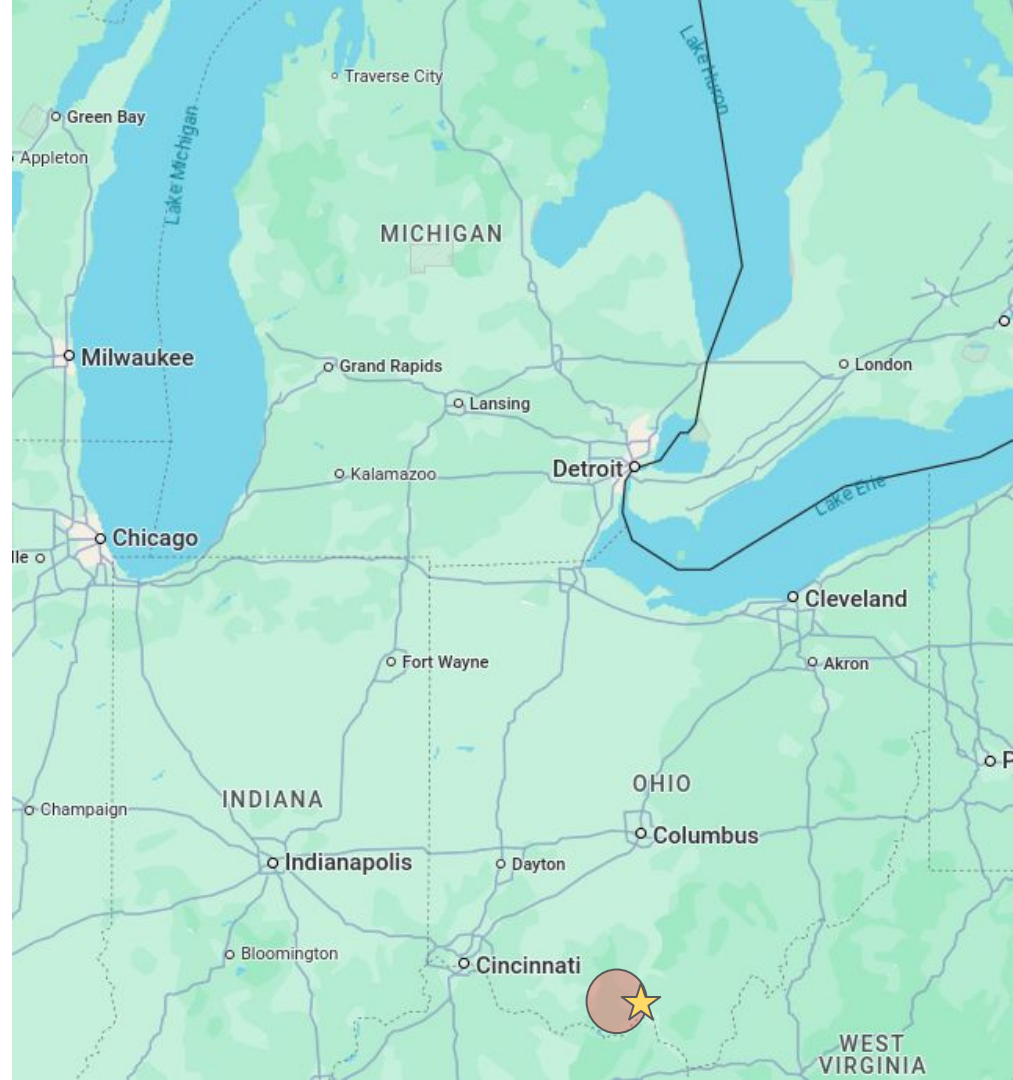
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Undergrad: Shawnee State Univ.

- Computer Engineering
- Game Programming



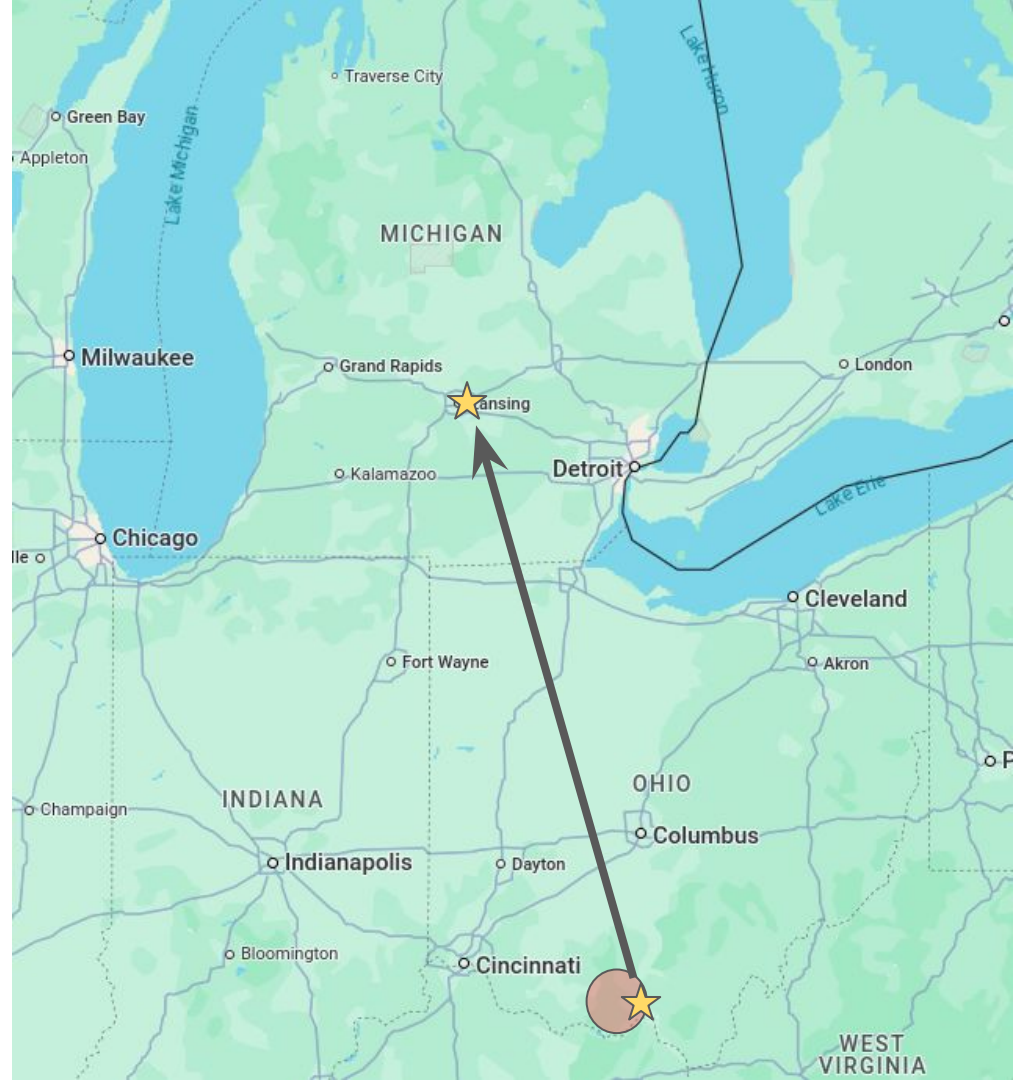
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- Computer Science
- Ecology, Evolution, and Behavior



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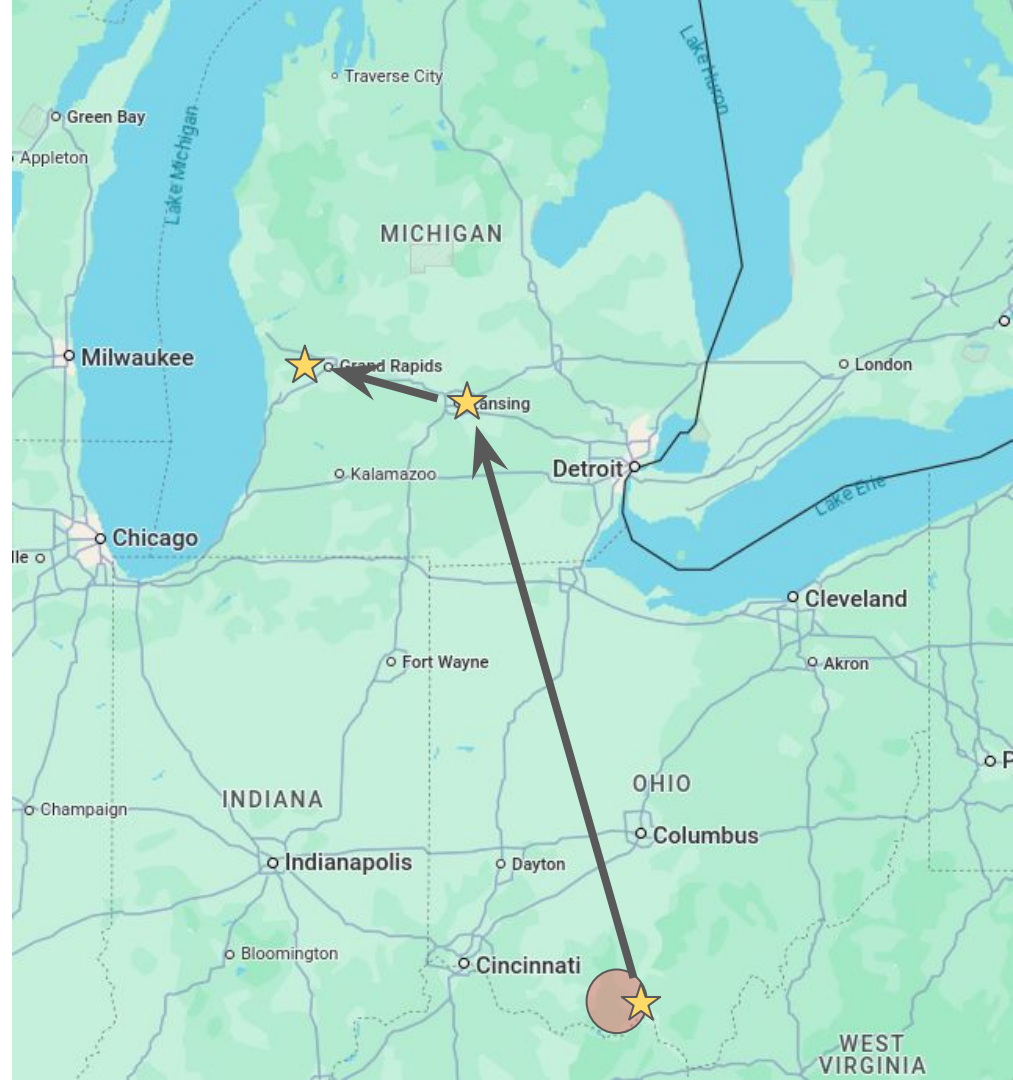
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Now: Here!



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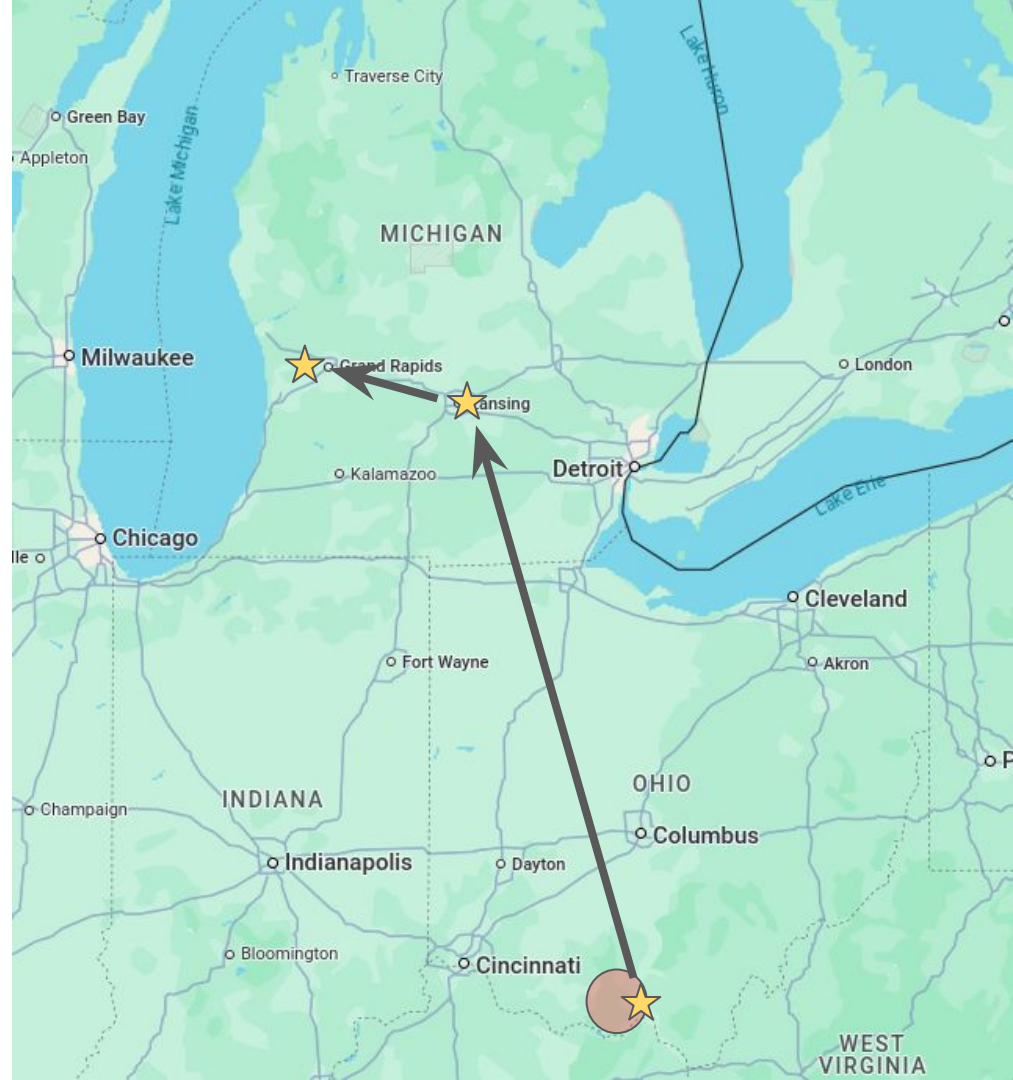
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???

Now: Here!



My research

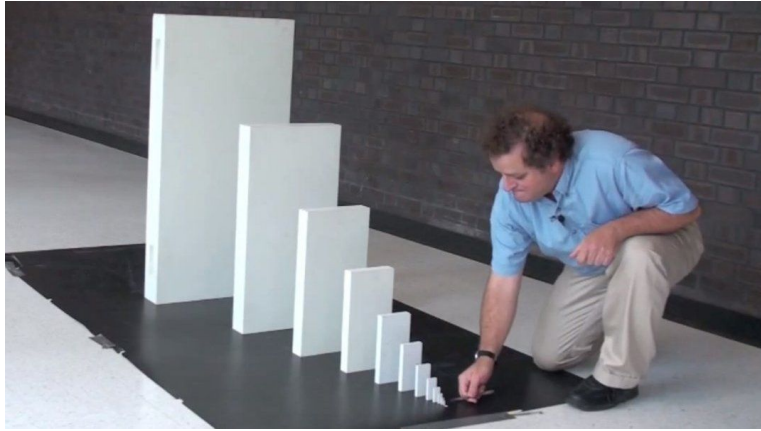
My research (10k-foot view)

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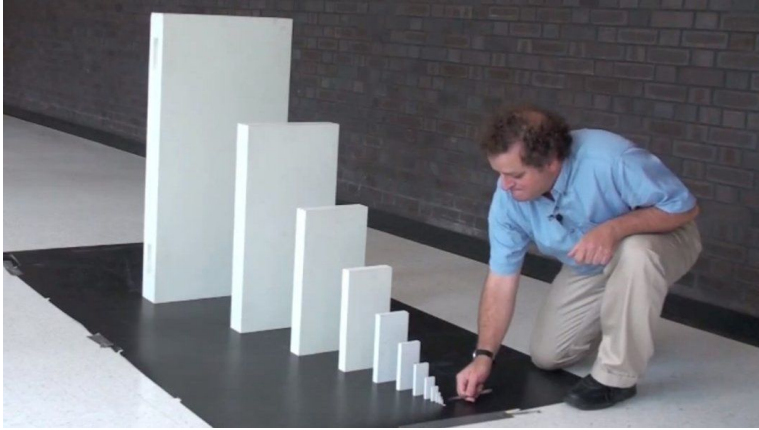
I use computational models...

... to study evolutionary biology theory

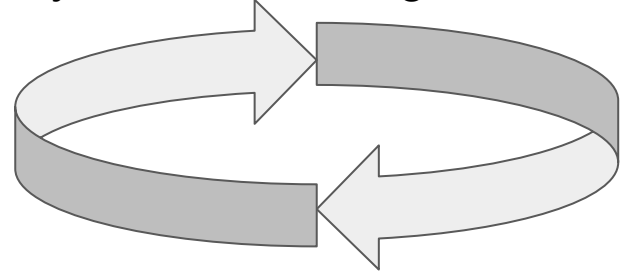
Historical contingency in evolution



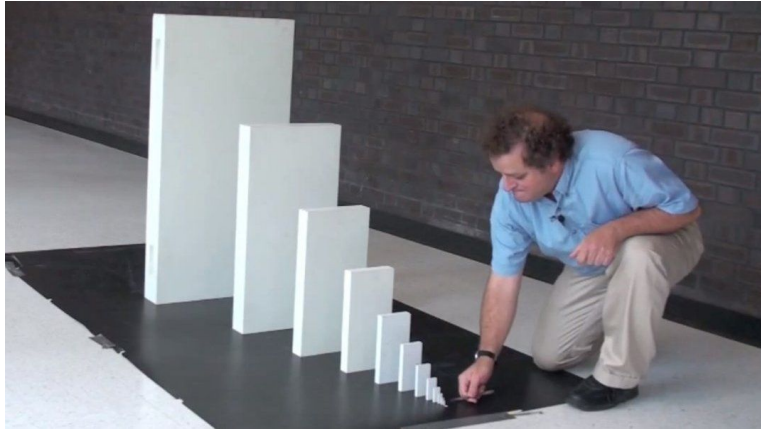
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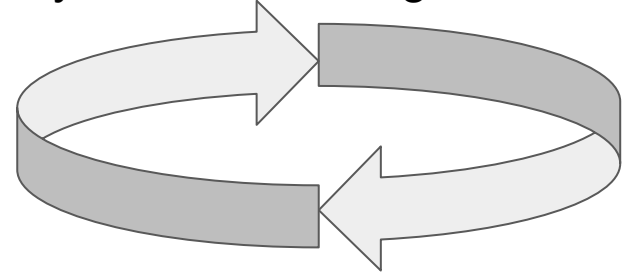
Evolvability - "how well" organisms evolve



Historical contingency in evolution



Evolvability - "how well" organisms evolve



Evolutionary dynamics of spreading invasive species



[Michael Montgomery - USDA FS](#)

Course logistics!

Course logistics



Course logistics

- Grades
- Links to other platforms



Course logistics



- Grades
- Links to other platforms

- Assignments



Course logistics



- Grades
- Links to other platforms

- Assignments



- Later in course
- Version control



Course logistics



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- Communication



- Later in course
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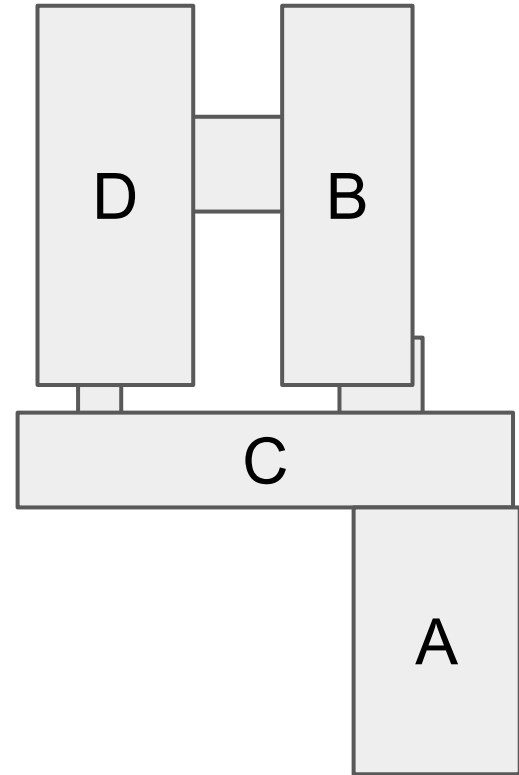
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Or email if private

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“Office” hours*:



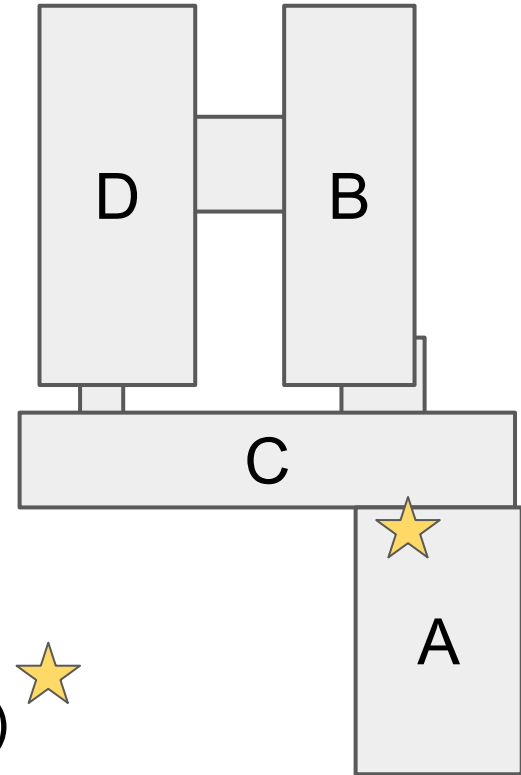
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“Office” hours*:

- Student Success Center (MAK A-1-101)
 - Tue. 10-10:50am



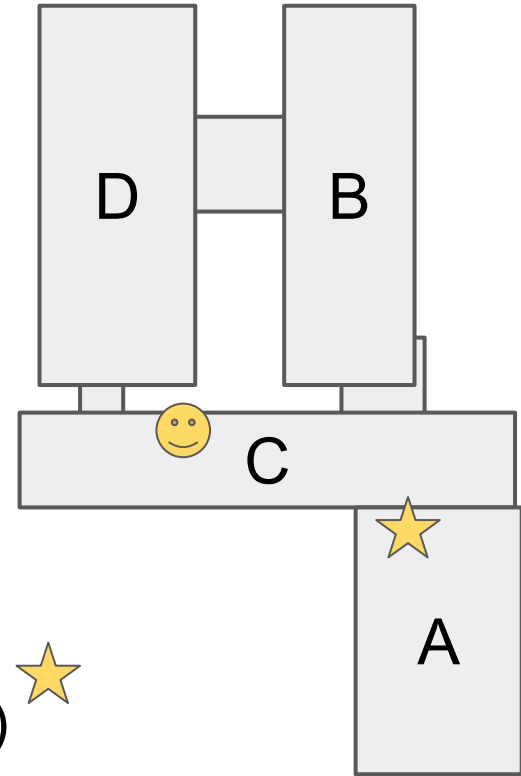
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“Office” hours*:

- Student Success Center (MAK A-1-101)
 - Tue. 10-10:50am
- My office (MAK-C-2-216)
 - Wed. 1-1:50pm



Textbooks

None required!

Syllabus has links (**FREE** from GVSU library) for extra reference

Grades 🙈

Grades 🙄

Assignment type	# of assignments	Total % of final grade
Check-in assignments	~ 11	25%
Projects	~ 4	30%
Command line quiz	1	5%
Midterm exam	1	20%
Final	1	20%

Late penalty: 10% per day late, up to 3 days

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Grades 🙈

Wednesday deadlines?

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Grades 🙈

A	94%	C	74%
A-	90%	C-	70%
B+	87%	D+	67%
B	84%	D	60%
B-	80%	F	Below 60%
C+	77%		

Grades 🙄

A	94%	C	74%
A-	90%	C-	70%
B+	87%	D+	67%
B	84%	D	60%
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C+	77%		

Final grade is rounded (ceiling function)
93.05% -> 94%

Other policies

- SAR Accommodations - happy to help, just reach out ASAP!
- Religious observance - let me know beforehand!
- Collaboration
 - Collaboration welcome on check-ins, but all work should be your own
 - Projects should be kept to conceptual questions (do not share code!)
 - Exams are NOT collaborative

Other policies

- Academic misconduct and plagiarism
 - Don't do it!
 - I take this stuff seriously
 - All text **MUST** be your own
 - You need to be able to explain **ALL** of your code!
- Artificial intelligence (AI)
 - Use it *sparingly*
 - You must disclose *why* you used it and give transcripts

Other policies

When in doubt, just ask! I'm here to help!

Extension / late days are way better than misconduct!!!

GVSU has resources!

- [Computing Success Center](#)
- [Replenish - Basic needs](#)
- [GVSU Counseling Services](#)

A supportive peer community helps with success

Look for Computing-related student organizations!

▲
JOIN THE
GVSU CIS
COMPUTING
CLUB!

Participate in a variety of activities:

- Company Visits
- Staff and Student Presentations
- Research Opportunities
- Tech News
- Skill Workshops and Team Projects
- Career Preparation

COME TO OUR WEEKLY MEETINGS!

Where: MAK A-1-105 (EOS Lab)
When: Thursdays 6 PM - 7 PM

Join on LakerLink! Find us on Discord!

Or contact us by email: cclub@mail.gvsu.edu



wic@mail.gvsu.edu



Hackers Analyzing
Threats (HAT) -
vijay_bhuse@gvsu.edu

Grand Valley Competitive Programming Club
(CPC) – woodriir@gvsu.edu

A note on transparency

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- I'm new(ish)!

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- I rely on your feedback

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 - For this semester
 - For future semesters

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Thanks in advance! :^)

Grading?

Communication?

Questions about logistics?

Content?

Resources?

A quick aside...



[NOAA - Unsplash](#)

A quick aside...



[Olivia Kudriavtseva - Unsplash](#)

A quick aside...

How many rounds of rock-paper-scissors
would it take to crown one champion?



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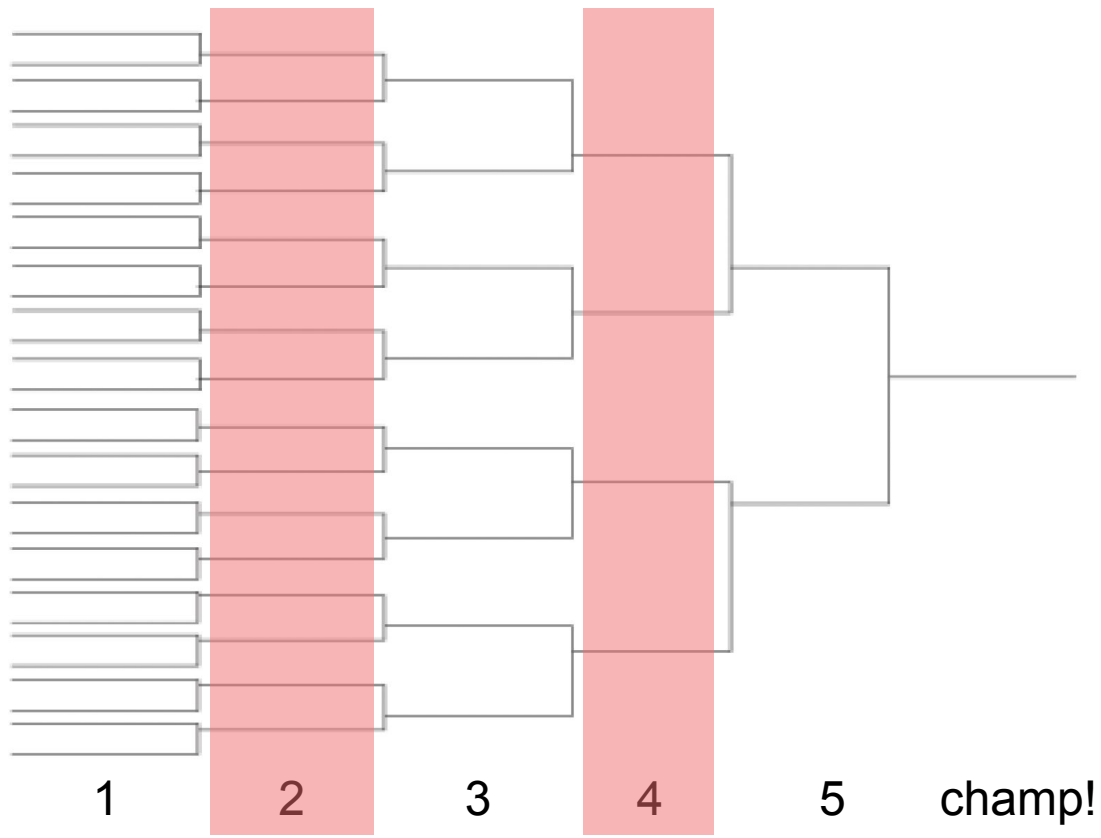
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[Olivia Kudriavtseva - Unsplash](#)

Rules:

1. No cheating (mind games are okay)
2. You must exchange names first
3. Always play best two of three!



[Olivia Kudriavtseva - Unsplash](#)

Back to CIS 241...

The big picture

Basic Linux terminal usage

```
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.8.0-40-generic x86_64)

System information as of Sun Aug 25 02:15:07 PM EDT 2024

System load:  0.13           Temperature:    126.0 C
Usage of /:   53.6% of 97.87GB Processes:      327
Memory usage: 17%           Users logged in: 0
Swap usage:   0%            IPv4 address for enp1s0: 35.39.29.66
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```
EOSLab
computing
```



```
-----
Contact HPC Support for help/questions
E-Mail      : arcit@gvsu.edu
-----
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Please do not share your login
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fergusa@eos01:~$ echo $1
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The big picture

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git basics
(version control)

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Why should I care? 🤔

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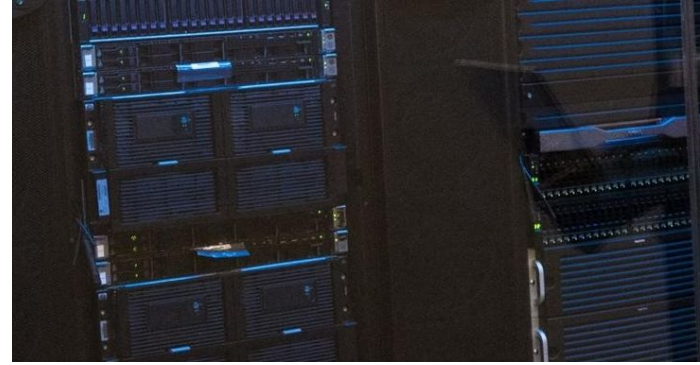
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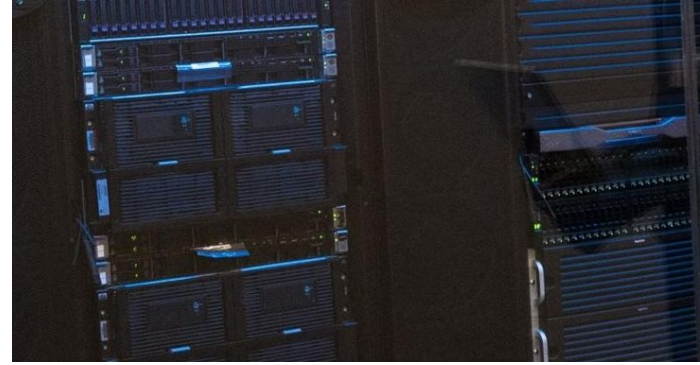
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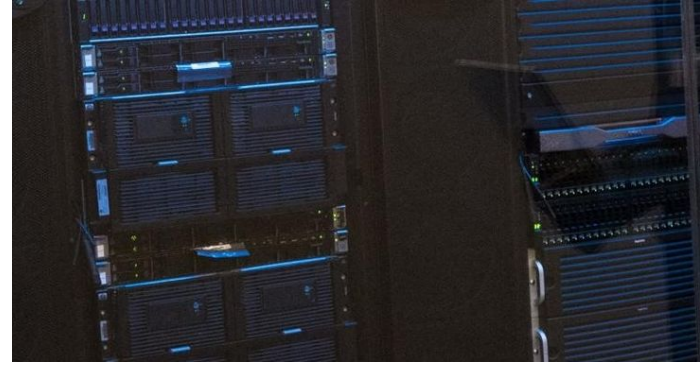
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- Efficiency



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Terminal:

- Accessing servers and HPC systems
- Efficiency
- Running programs without GUIs

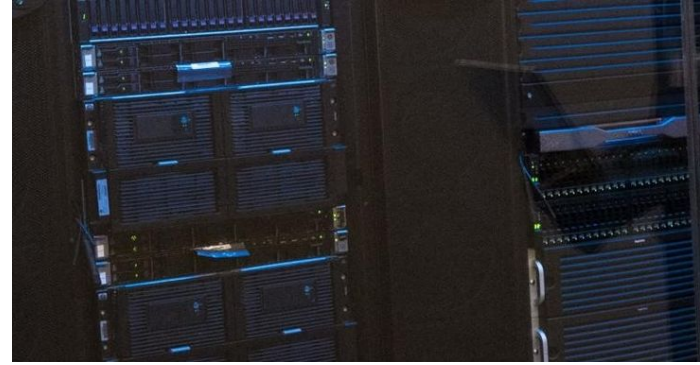


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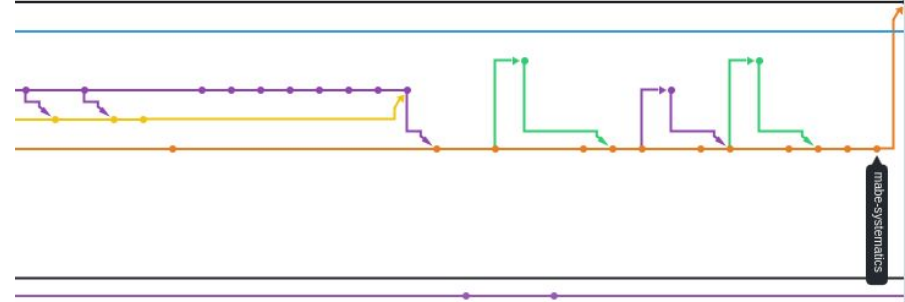
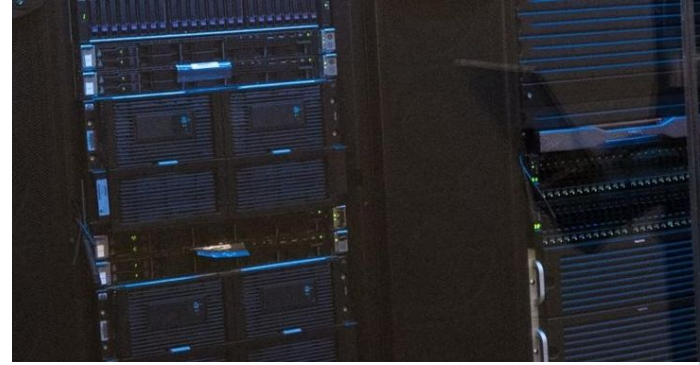
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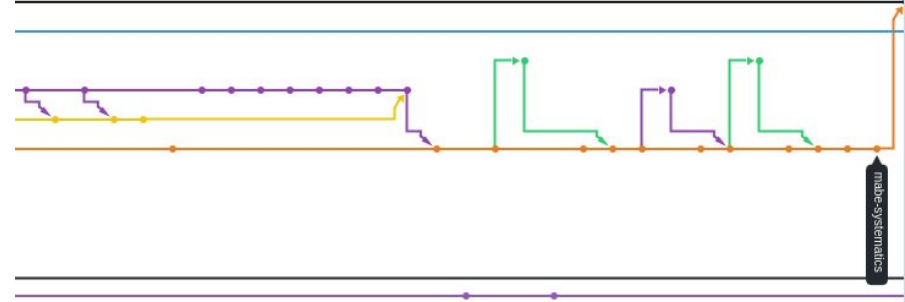
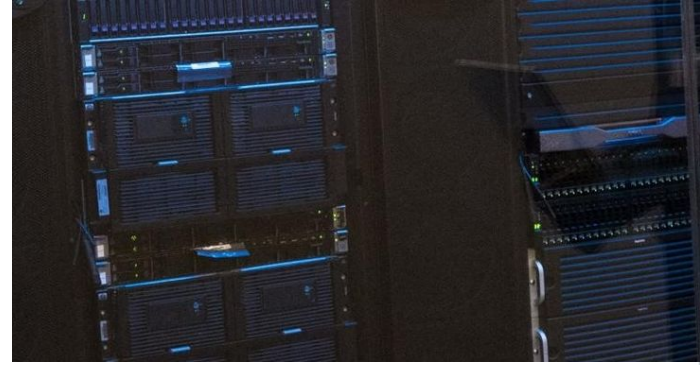
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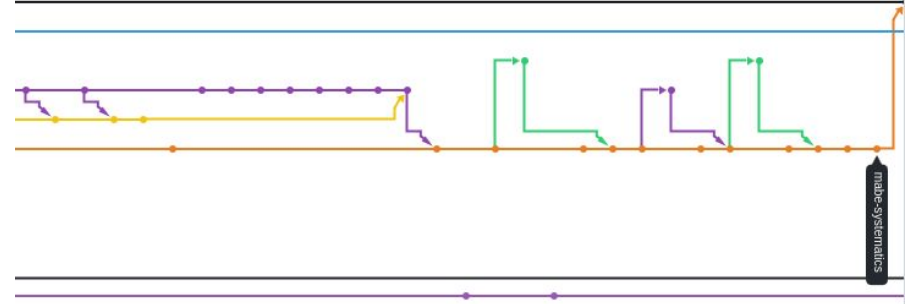
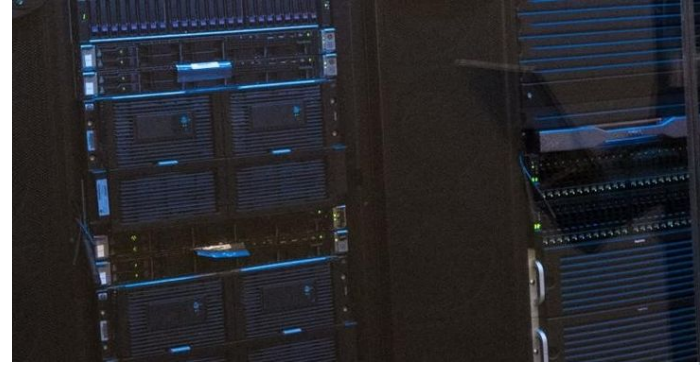
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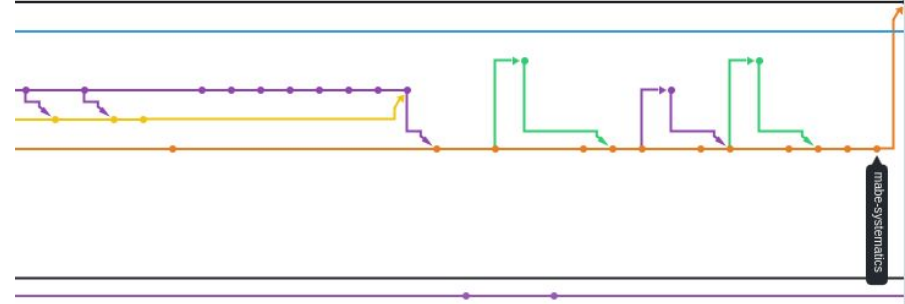
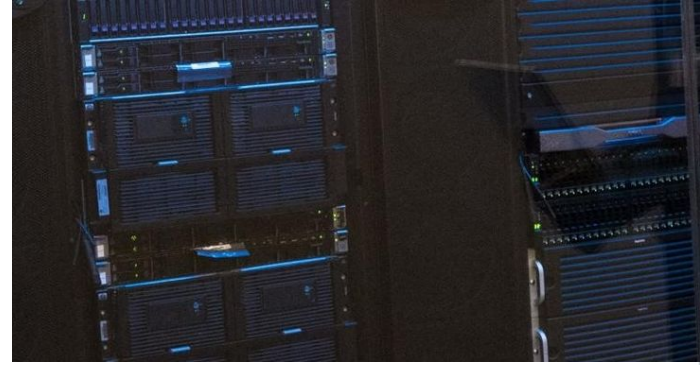
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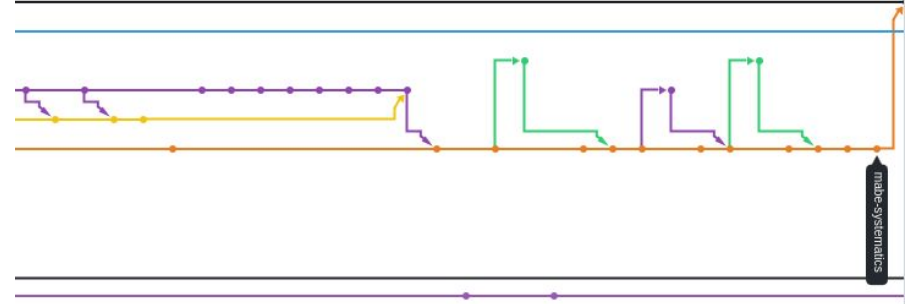
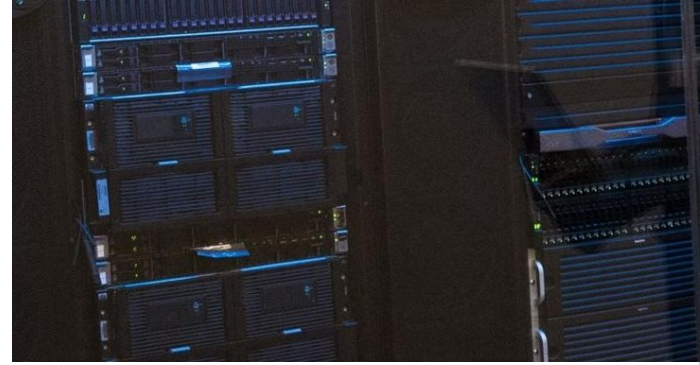
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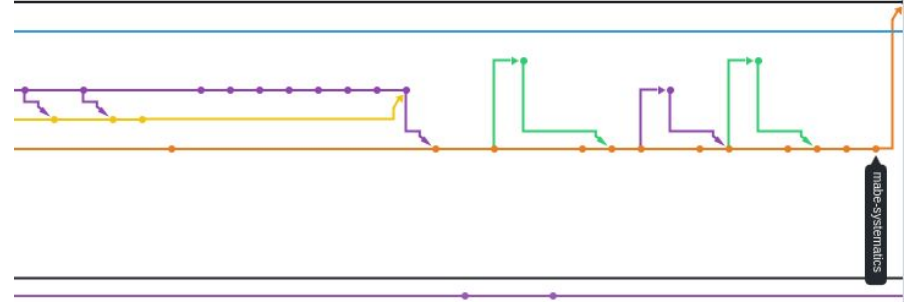
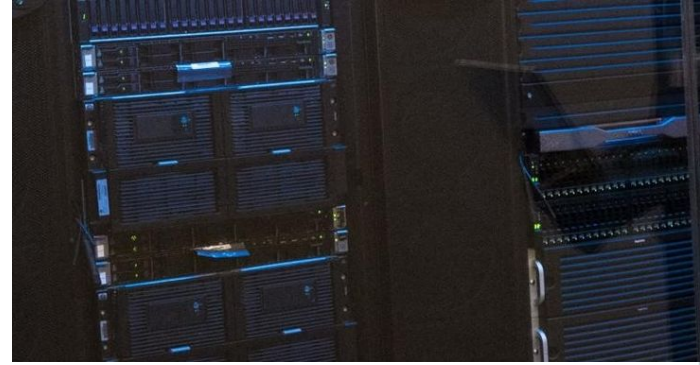
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- Will improve your high level programming



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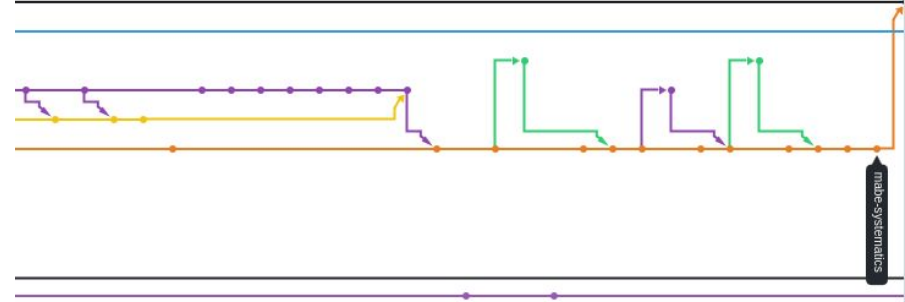
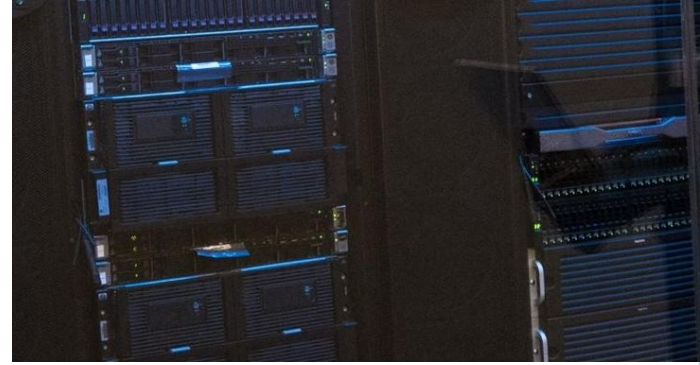
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- Used everywhere, collaboration
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C:

- Understand the low level (memory management)
- Will improve your high level programming
- Still used in embedded systems



[SparkFun - \(Wikimedia\)](#)